Exploring a Virtual Pet as an Adaptive Agent in Personalized VR Rehabilitation: A Country Fair Serious Game for Stroke Survivors

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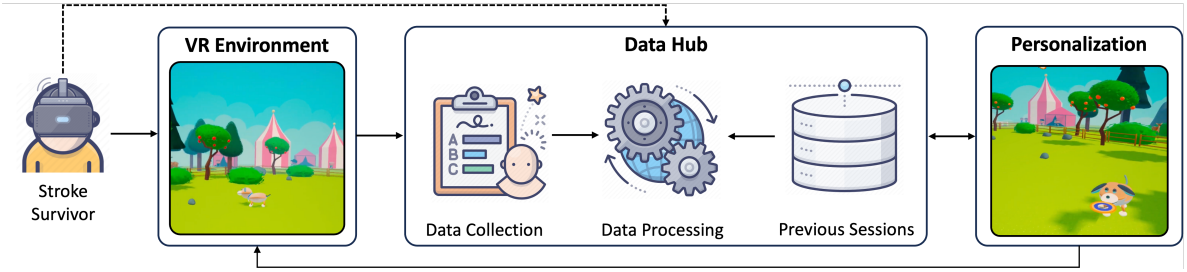


Figure 1: Overview of the VR-rehabilitation application, with a stroke survivor interacting with a country fair environment while performance data is collected and processed in a data hub to drive personalized adaptation of the virtual pet–mediated gameplay.

Abstract

Traditional stroke rehabilitation methods fail to sustain long-term motivation due to their repetitive and non-adaptive nature. This work presents a Virtual Reality (VR) country fair serious game designed to support stroke survivors during upper-limb rehabilitation through personalized and engaging interactions. The game integrates a virtual pet as the central engagement mechanism. Survivors complete frisbee-throwing tasks toward predefined targets in the virtual environment, after which the pet retrieves the object and returns it, creating a continuous interaction loop. Visual and auditory feedback, including celebratory cues such as barking, reinforces progress and maintains attention. A personalization module continuously collects performance data and dynamically adjusts target placement according to user's ability, ensuring an appropriate challenge level while reducing monotony. The game was evaluated with 38 participants, who reported positive feedback, describing it as fun, easy to use, and appropriately challenging. Results suggest that adaptive personalization enhances motivation and provides useful behavioral data for new features.

Index Terms: Virtual Reality, Virtual Pet, Stroke Rehabilitation, Personalization, Country Fair, Serious Game

1 Introduction

Stroke affects millions of individuals each year, resulting in long-term motor impairments. [6, 4, 5]. Yet, traditional methods lack individualization, applying the same therapeutic approaches across survivors despite substantial differences in severity, recovery stage, and personal characteristics. This uniformity can reduce engagement and limit long-term adherence to rehabilitation programs [8, 2]. Hence, there is a growing need for personalization methods that adapt rehabilitation activities to each individual's capabilities, ensuring an appropriate level of challenge while maintaining motivation and supporting continuous progress [3, 7].

This work presents a Virtual Reality (VR) serious game for stroke survivors upper-limb rehabilitation, framed within a country fair–themed approach. The game transforms repetitive rehabilitation exercises into engaging, goal-oriented activities through a virtual pet–driven gameplay loop in an immersive environment. A key contribution is a continuous personalization module that adapts task difficulty based on survivor performance, enabling individualized progression over time. The game incorporates multimodal feedback to reinforce actions and sustain engagement.

2 Methods and Materials

The design followed a Human-Centered Design (HCD) methodology with stroke survivors and healthcare professionals from a rehabilitation center. Figure 1 illustrates the VR-based personalized country fair approach [1, 3]. The **VR environment** represents the country-fair world where rehabilitation activities take place. Survivors throw virtual frisbees toward moving targets distributed across the fairground, requiring coordinated upper-limb movements, aiming, and timing. After each throw, a **virtual pet** retrieves the frisbee and returns it to the survivor, acting as the central interaction mediator and reinforcing a continuous, engaging feedback loop that encourages sustained participation. The pet also provides expressive feedback during successful actions, strengthening user engagement and emotional connection to the activity. This design supports rehabilitation objectives by promoting arm strength, range of motion, and movement precision.

The **Data Hub** is the analytical core of the personalization process. It aggregates gameplay data, including movement accuracy, reaction speed, task success, interaction patterns, and historical session data, to extract meaningful indicators of performance and progress. The Data Hub comprises three main modules: *data collection*, which gathers metrics from the VR environment and user interactions; *data processing*, which applies a rule-based dynamic difficulty adjustment approach to adapt rehabilitation tasks while maintaining an appropriate level of challenge; and *historical data access*, which compares the survivor's current performance with previous sessions to identify progress trends, support long-term personalization, and detect signs of fatigue or performance decline.

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The **Personalization Module** uses the insights generated to adapt game parameters and tailor each activity to the survivor's evolving needs. This includes adjusting difficulty, target size, distance, speed of moving targets, and the type of feedback provided. The updated configurations are then applied to the VR-based experience, ensuring that rehabilitation remains appropriately challenging, engaging, and aligned with the survivor's abilities, thereby supporting continuous progression and sustained motivation.

The game was development for the Meta Quest 3 using the Unity game engine with C# scripts. The scene integrates original 3D assets modeled in Blender, including elements such as trees, bushes, the dog, bow, and frisbee, together with complementary low-poly assets obtained from Poly Pizza under open licensing, including circus tents, pedestrian characters, and a Ferris wheel. Spatial audio was implemented through FMOD Studio to provide directional auditory cues and dynamic sound adaptation, including pitch and volume variation to reduce repetition and strengthen immersion.

3 User Study and Results

To evaluate the personalization features and the virtual-pet integration, a user study with healthy individuals was conducted, allowing validation before moving into the stroke population. Participants were introduced to the objectives, tasks, and overall experimental setup and design. After providing informed consent, they filled a questionnaire regarding background (e.g., demographic data, experience with VR technologies). At the eng, they filled out a post-study questionnaire covering their experiences, preferences and suggestions. Data were collected using a Likert-type scale (1 – Low, up to 7 – High) with open-ended questions to capture both quantitative ratings and qualitative feedback. The study concluded with a follow-up interview to collect qualitative feedback.

Thirty-eight participants tried the game (6 female), with ages ranging from 21 to 57 years (M=25.2). The sample included participants from diverse backgrounds, including healthcare professionals as well as individuals with expertise in human-computer interaction, bio-informatics, game design and development, and VR, contributing to an heterogeneous perspective on the subject being handled. Overall, the game was considered intuitive and enjoyable to use. Participants appreciated the cartoon-like visual style, which created a light and approachable atmosphere, and several suggested adding new fair activities to further expand the experience. Regarding the virtual pet, participants particularly enjoyed interacting with it, reporting that it helped distract from the repetitive nature of upper-limb exercises. Many suggested richer interactions (e.g., commands such as sit or jump) and additional personalization options, including selecting the pet's breed, appearance, or interaction style. Participants also responded positively to the personalization mechanisms. They noticed the dynamic adaptations (*Mdn = 6*), perceived them as beneficial (*Mdn = 6*), and reported increased engagement (*Mdn = 6*) and motivation (*Mdn = 5.5*). They trusted the system to adapt the rehabilitation appropriately (*Mdn = 6.5*) without making the session overly tiring (*Mdn = 6*), while also expressing willingness to continue using such a personalized XR rehabilitation system in future sessions (*Mdn = 5.5*). Although participants generally understood the adaptive behavior (*Mdn = 5*) and found the personalized feedback useful (*Mdn = 5.5*), these aspects were also identified as opportunities for future improvement.

Overall, these findings indicate a positive perception of the virtual pet and the personalization approach. The combination of adaptive difficulty and an engaging companion has the potential to increase motivation, and make repetitive exercises more enjoyable while maintaining an appropriate level of challenge. Notwithstanding, these results should be interpreted as an initial validation with a non-clinical population. Future studies involving stroke survivors are required to evaluate the clinical effectiveness, long-term engagement, and therapeutic benefits of the proposed approach.

4 Conclusion and Future Work

This work presents a VR-based country fair serious game designed to support stroke survivors during upper-limb motor recovery. It is characterized by its emphasis on personalization and the integration of a virtual pet as a central interaction and engagement mechanism. The game was evaluated with 38 participants, yielding positive feedback. It was both fun and easy to play, while also being challenging due to its continuous personalization module, which helps maintain their motivation.

Moving forward, we intend to expand the interaction repertoire of the virtual pet by introducing new actions and behaviors that promote richer and more engaging dynamics, thereby supporting a broader range of motor and cognitive stimulation during rehabilitation. Furthermore, we plan to conduct a longitudinal study in a rehabilitation center to systematically evaluate the impact of the virtual pet on engagement, adherence, and motor recovery outcomes over time. In this vein, we also consider incorporating multiple types of virtual pets to compare stroke survivors' preferences and perceived engagement, as well as to investigate how different pet characteristics influence motivation and rehabilitation performance.

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